

RAMITH K S

Senior Backend Engineer

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years of experience building scalable Python/FastAPI services, distributed systems and cloud-native applications. Experienced in developing AI-enabled backend features using LangChain, LLM APIs, RAG pipelines and lightweight machine-learning models.

PROFESSIONAL EXPERIENCE

Thermo Fisher Scientific

Senior Backend Engineer

Feb 2026 - Present

- Build Python/FastAPI services for high-reliability e-commerce checkout and order-processing workflows on Kubernetes.
- Implemented Kafka-based acknowledgement processing, validation and status updates across asynchronous order workflows.
- Improved reliability through structured error handling, retries, observability and integration testing.

Zolnoi Innovations

Backend Engineer

Oct 2024 - Present

- Built industrial IoT analytics APIs using FastAPI, PostgreSQL, Redis and MongoDB, processing 10,000+ daily requests.
- Integrated rule-based and ML-generated anomaly insights into production, energy and cycle-time analytics workflows.
- Designed a multi-channel alerting system with retries and failure handling, processing 5,000+ notifications daily.

JustPoll (Streetlab Internet Pvt Ltd)

Founder & CTO

Jul 2021 - Aug 2024

- Founded and scaled a social platform to 1,500+ Android and 600+ iOS users; architected TypeScript backend and Flutter app.
- Boosted query performance by 40% via schema optimization; cut deployment time by 60% with ECS autoscaling and CI/CD pipelines.
- Led 5-member engineering team through 20+ production releases; improved user retention by 65% through mentoring and architecture reviews.

TECHNICAL SKILLS

Backend & Languages

Python, FastAPI, AsyncIO, REST APIs, Node.js, TypeScript

Generative AI & Machine Learning

LangChain, LangGraph, LLM API Integration, RAG, Prompt Engineering, Structured Outputs, Embeddings, scikit-learn, Model Serving

AI Data & Storage

pgvector, Vector Search, PostgreSQL, Redis, MongoDB, Pandas, NumPy

Systems & Cloud

Microservices, Kafka, Event-Driven Architecture, Kubernetes, Docker, AWS, GitHub Actions, CI/CD, Observability

Testing & Reliability

Pytest, Integration Testing, Retry and Fallback Workflows, Root Cause Analysis

Additional

React, Flutter

EDUCATION

Bachelor of Computer Applications

Christ University, Bengaluru
2018 - 2021 | CGPA: 3.3 / 4.0

RECOGNITION

- HackBangalore Winner 2024
- AWS Cloud Foundations Certified
- Open Source Contributor

KEY PROJECTS

AlgoHabit — AI-Powered Learning Platform [React, TypeScript, LangChain, RAG, pgvector, OpenRouter AI, Supabase, Razorpay, PWA]

- Built and launched AlgoHabit, an AI-powered DSA learning platform designed around habit formation, streaks, and gamified progression with 50+ active users.
- Engineered React PWA achieving 95+ Lighthouse performance score with offline-first architecture using service workers for 100% offline functionality.
- Built a LangChain-based LLM pipeline using OpenRouter, prompt templates, structured output parsing, retries and model fallback to generate personalised learning plans.
- Implemented a RAG pipeline using document chunking, embeddings and pgvector-based semantic search to generate syllabus-grounded learning recommendations.
- Deployed Supabase backend with Row-Level Security (RLS) securing 400+ user records; enabled zero-downtime Razorpay payments via serverless edge functions.
- Implemented gamification engine tracking 10,000+ XP transactions, improving 30-day retention by 60%.

Ledgerly — Smart Finance Tracker SaaS [FastAPI, React, PostgreSQL, scikit-learn, MLflow, statsmodels, Pytest]

- Built scikit-learn pipelines using TF-IDF and Logistic Regression for transaction classification, statistical forecasting for spending trends and MAD-based anomaly detection.
- Served ML predictions through FastAPI endpoints with Pydantic validation, model loading, logging and health checks.
- Used MLflow for basic experiment tracking, model metrics and version comparison.

HelpMyBuddy — HackBangalore Winner 2024 [Node.js, TypeScript, FastAPI, React Native, MongoDB, PostgreSQL]

- Built real-time geo-based emergency help marketplace connecting users to nearby helpers for breakdowns and medical emergencies.
- Designed a low-latency geospatial matching engine using MongoDB spatial queries, achieving <500ms matching within a 5 km radius.
- Developed React Native (Expo) mobile app enabling live location sharing, helper discovery, and emergency request workflows.
- Built TypeScript-based Node.js + Express backend to handle authentication, matching, and request orchestration with FastAPI-powered ML services.

AG-Wallet — Agricultural Supply Chain Platform [Android (Java), MySQL, Google Maps API]

- Built Android app connecting farmers to distributors with real-time price tracking and downstream visibility.
- Integrated Google Maps API for location-based discovery and rating system for distributor evaluation.
- Enabled transparent agricultural supply chain with location-based discovery and quality ratings.